AKANKSH CAIMI

MSc Applied Artificial Intelligence

Applied AI Engineering · Graduate Software Engineering

Coventry, UK · +44 7352 691890 · akankshcaimi@gmail.com
github.com/AkankshCaimi · linkedin.com/in/akanksh-caimi

Completing an MSc in Applied Artificial Intelligence at Warwick, researching when language models should answer or abstain. Computer science graduate with experience developing SQL fraud-detection tools and Python automation at American Express. Builds across machine learning, full-stack applications and systems programming, with an emphasis on evaluation and inspectable results.

EDUCATION

University of Warwick | MSc Applied Artificial Intelligence

Sep 2025 – Sep 2026 · Predicted Merit (65%) · WMG Excellence Scholar

Dissertation submitted September 2026. Study includes deep learning, reinforcement learning, data mining and AI for industry.

IIT Ropar | B.Tech Computer Science and Engineering

Nov 2021 – May 2025 · GPA: 7.0/10

Foundation in software engineering, data structures, algorithms, databases and artificial intelligence.

TECHNICAL SKILLS

Languages: Python, C/C++, JavaScript, SQL.

Applications: React, React Native, Flask, Django, PostgreSQL, SQLite, WebRTC.

AI & delivery: PyTorch, TensorFlow, Hugging Face, scikit-learn, pandas, NumPy, OpenCV, Git, Docker, pytest, GitHub Actions.

APPLIED AI RESEARCH

Threshold-conditioned LLM abstention | MSc dissertation

Mar – Sep 2026 · Python, PyTorch, Hugging Face, LoRA · Code · Dissertation

- Developed self-sampled confidence training with one LoRA adapter per model, allowing a prompt threshold to control answering versus abstention across Mistral-7B, Llama-3.1-8B and Gemma-2-9B.
- Evaluated on 500 held-out TriviaQA questions. For tuned Mistral, raising the threshold from 0 to 0.9 reduced incorrect answers from 146 to 15, while answer coverage fell from 97.8% to 42.6%.
- Analysed the accuracy–coverage trade-off with paired tests; a separate SimpleQA evaluation exposed near-total abstention at higher thresholds, defining a limit to generalisation.

Related research: sole-author manuscript, “Diagnosis of Reward-Based Hallucination Mitigation in Large Language Models” (2026). · Manuscript

ENGINEERING EXPERIENCE

American Express | Data Analyst Intern

May – Oct 2024 · Gurugram, India

- Designed and deployed SQL scanners to identify fraud patterns in customer transaction data, translating findings into recommendations for fraud operations stakeholders.
- Built Python utilities for data preparation, analysis and visualisation, saving 10+ hours of manual work per week; tools were adopted by the wider team for faster error reporting.

ENGINEERING EXPERIENCE / CONTINUED

Ambiente Technologies | Software Development Intern

May – Jul 2023 · Mohali, India

- Developed a Python application using OpenCV and dlib to automate identity matching that previously relied on manual lookup.
- Built WebRTC calling features including audio/video controls, screen sharing and recording; configured STUN/TURN connectivity for calls across networks.

SELECTED ENGINEERING PROJECTS

Community-centre booking application | In development

Oct 2026 · React, Flask, PostgreSQL, Docker · Code

- Developed booking validation and administrative approval flows, with scheduled jobs for automatic declines and data retention.
- Added backend and frontend tests, linting and container builds to GitHub Actions; CI runs database-backed tests against a fresh PostgreSQL instance.

ZAPPFIX service marketplace | Team project

Jan – May 2024 · React Native, Django, PostgreSQL · Code

- Contributed to a mobile marketplace for everyday services, with customer/provider profiles, location-aware service discovery, job requests, status tracking and reviews.

Hostel management portal | Team project

Jan – May 2024 · React, Django, SQLite · Code

- Implemented React screens and API integration for student profiles, application status/details, administrator views, room inspection and payment-proof workflows.

32-bit RISC-V simulator | Four-person team project

C/C++, Python · Code

- Contributed to a simulator spanning single-cycle and pipelined execution, instruction/data caches and a Python GUI for inspecting processor state.

FURTHER MACHINE LEARNING WORK

- Chest X-ray classification: benchmarked a four-block CNN against VGG16 with two-phase fine-tuning; used Grad-CAM and SHAP to inspect model behaviour. · Code
- Diabetes risk classification: compared supervised models on CDC survey data and examined class imbalance through per-class recall, alongside exploratory clustering. · Code

AWARDS & CONTRIBUTION

Warwick Award (Postgraduate): 100 hours of extracurricular activity. AI Hackathon, IIT Ropar: first runner-up. AWS Academy Machine Learning Foundations: 20-hour course, February 2026.

Contributed 35 hours through Warwick STAR and school volunteering, supporting English learning for refugees and asylum seekers and GCSE revision at West Coventry Academy.